Plastic Object Detection by Using Computer Vision and AI Techniques

Shubham Santosh Auti[1] and Dr. Jyoti Vishnu Joglekar[2]
[1] Department of Computer Engineering, KJ Somaiya College of Engineering, Mumbai - 77, Maharashtra, India
shubham.auti@somaiya.edu
[2] Department of Computer Engineering, KJ Somaiya College of Engineering, Mumbai - 77, Maharashtra, India.
jyoti.joglekar@somaiya.edu

Abstract. Recent years have seen the production of a broad range of polymers by the petrochemical sector which have tremendously helped human development. On the other hand, improper disposal of plastic has gravely harmed ecosystems, animals and resource sustainability by means of pollution. Given its long decomposition time and consistent character plastic waste management is a serious global problem. Recycling helps to reduce virgin plastic production, lessen environmental impact and support circular economy initiatives. Still manual plastic waste sorting is physically taxing and economically unproductive. Using hyperspectral photography, the proposed method accurately identifies and separates plastic waste. Initially, it separates plastic waste from regular waste in disposal sites using object recognition and hyperspectral image classification. After that, the separated waste is handled by a multi-layered sensor array designed to identify various plastic materials depending on unique spectral fingerprints. This approach exactly separates recyclable materials from non-recyclable polymers. While recyclable plastics are further categorized by type for efficient processing, non-recyclable plastics are under control to lower environmental damage. By means of reduction of their footprint, this system aims to increase recycling efficiency and minimize the environmental impact of plastic waste by using an automated, scalable and efficient framework for plastic waste management.

Keywords: Object Detection, Computer Vision, Artificial Intelligence, Plastic material, Recyclable Plastic

1 Introduction

Since plastics break down slowly and cause enormous environmental problems like pollution, damage to animals and dangerous chemical leakage indicates that the plastic waste management is a great international environmental issue. Dealing with these issues calls both lowering the production of new plastic and increasing the efficiency in recycling techniques. since manual sorting techniques are labour intensive, expensive and worthless considering the complexity and spectrum of plastic debris which leads to a great market for automated systems which are scalable and highly efficient for plastic identification before separation.

While present automated waste management systems focus on broad waste categories, they lack the accuracy needed to identify particular plastic types thereby allowing PET, HDPE and LDPE to be differentiated from non-recyclable polymers. The suggested work presents computer vision methods by extracting whole spectrum attributes of materials using hyper-spectral photography. This suggested method allows excellent identification of plastic material by use of their special reflectance and absorption characteristics therefore optimizing segregation of plastic. A basic feature of this kind of technology that increases location-based waste collecting and operational efficiency is geo-tagging areas of interest inside trash disposal sites. Moreover, the degree of contamination in mixed plastic waste is determined by the accuracy of hyperspectral imaging, thereby simplifying early recycling procedures as shredding and washing. This approach provides most efficient way of treatment by maximizing the recycling flow.

By reducing reliance on hand sorting, improving recycling efficiency, and adopting a circular economy whereby resources are once more used to cut waste, the project improves environmental sustainability. Reducing the consequences of plastic waste supports worldwide initiatives aiming at climate change as well. Combining hyperspectral photography, artificial intelligence categorization, and spatial analysis, the work offers in general a transforming answer to plastic waste management. Early recycling problems should be addressed to boost efficiency, accuracy, and scalability, thereby encouraging creativity in waste management and allowing the path for a better, sustainable future.

2 Related Work

The work in [1] provides information about the use of cutting-edge machine learning techniques especially Random Forest classifiers to identify and categorize litter in aerial photographs taken by remotely piloted aircraft systems (RPAS). With great accuracy exceeding 0.88 in both test and validation datasets the proposed method efficiently identified plastic debris and distinguishes it from commodities including wood, glass, and soil.

The work in [2] looks at hyperspectral imaging as a efficient approach for detection and classification of aquatic environment floating plastic waste. The work precisely identified plastic debris, plants and water using hyperspectral sensors (FX10 and FX17) spanning a spectral range of 400–1700 nm. While preprocessing processes including converting raw data to relative reflectance and intensity normalisation, enhanced signal to noise ratios so boosting material classification dependability controlled experimental settings guaranteed accuracy. By means of crucial absorption peaks in the NIR - SWIR region, namely 1210 nm, 1430 nm, 1150 nm, and 1450 nm, linear discriminant analysis (LDA) was applied to categorize materials including HDPE, LDPE, PET, PS, and PP. In separating floating plastics from water and vegetation, hyperspectral imaging exceeded multispectral indices including FDI and NDVI. Sentinel-2 satellite bands (e.g., B4, B6, B8, and B11) were underlined for their relevance in remote sensing.

Using multispectral vision from UAVs, the study in [3] presents a multi-step method for identifying macro plastics in river ecosystems. Particularly at lower altitudes where greater spatial resolution improves accuracy the study obtains strong plastic garbage detection by using a cascade of two random forest classifiers. The work suggests adding further infrared (IR) spectral bands to use the special reflectance characteristics of plastics, therefore improving distinction from other materials and helping to overcome these obstacles. Deep learning with convolutional neural networks (CNNs) is one of the advanced machine learning approaches advised to increase adaptability to different environmental circumstances and visual variations.

This work presents Fourier Transform Infrared (FTIR) reflectance spectroscopy throughout the Near Infrared (NIR) and Mid Infrared (MIR) domains to investigate the identification of plastic resins. Although certain plastics found the NIR domain helpful, its dependability for thorough resin identification was compromised by unpredictability brought on by colour and shape.

Using the 955 to 1700 nm spectral region to precisely identify 13 plastic kinds via unique spectral signatures the work in [5] shows hyper-spectral imaging (HSI) as a transforming option for plastic waste management. Using the Savitzky-Golay filter and Principal Component Analysis (PCA) the study enhanced data clarity and classification accuracy so addressing issues including sample variability and noise from surface roughness. Using unsupervised machine learning including clustering cost-effective plastic categorization was made possible without large-scale labelled data. The practicality of including HSI into recycling processes was shown by the integration of a commercially accessible spectrometer therefore guaranteeing great accuracy with minimum data processing.

Especially under constraints of climate change and desertification, the study in [6] shows the need of spotting rich rangeland regions to support sustainable cattle husbandry and minimize overgrazing which can result in bare patches and alien species invasion. It shows how well high-resolution NDVI data can be used as a reasonably cheap technique to map spatial variability in productive regions and sheep grazing preferences in the semi-arid Upper Karoo ecosystem.

3 Implementation

3.1 Dataset

The Hyperspectral Plastics Dataset, supplementary to research article [2] https://data.4tu.nl/articles/dataset/Hyperspectral_plastics_dataset_supplementary_to_the_paper_Advancing_floating_plastic_detection_from_space_using_hyperspectral_imagery_/14518278, is used as the dataset for this proposed work for plastic detection using hyperspectral imaging techniques. Organized into Virgin Plastics and Riverbank-Harvested Micro litter, the dataset allows analysis of both pristine and real-world samples, fostering robust model development for diverse conditions, including varying material degradation and contamination levels. The image **Fig.1.1** [1] below shows us the sample data from the dataset.

Fig.1.1 different kinds of plastics

3.2 Dataset Levels:

3.2.1 Level 0 (Raw Hyperspectral Data):

Captured by specim FX10 (VIS-NIR: 400–1000 nm) and specim FX17 (NIR-SWIR: 900–1700 nm) cameras, offering extensive spectral coverage with resolutions of 5.5 nm and 8 nm, respectively. Includes white and dark reference files for calibration, image metadata for detailed experiment replication, and average capture images for quick visual validation.

3.2.2 Level 1 (Processed Data):

Provides relative reflectance and normalized intensity formats, with manually selected representative pixels for each plastic category. MATLAB-compatible datasets ensure usability in computational workflows, enhancing data standardization and comparison across conditions.

3.2.3 Level 2 (Averaged Spectral Data):

Summarized mean reflection spectra and standard deviations for eight plastic categories. Facilitates exploratory analysis and algorithm validation, offering concise statistical representations of complex hyperspectral data.

3.3 Plastic Item Data and Statistics:

An accompanying Excel file that provides Reflectance values and standard deviations for sampled plastic and Unique item IDs, descriptive labels, and image references to enhance traceability and usability.

3.4 Imaging System Details:

Specim FX10 Camera Visible - Near Infrared (VIS-NIR) which provides 1024 pixels per line scan, 224 wavelength bands, 5.5 nm spectral resolution.

Specim FX17 Camera Near Infrared – Short wave Infrared (NIR-SWIR) which provides 640 pixels per line scan, 112 wavelength bands, 8 nm spectral resolution

3.5 Data Cleaning

Data cleaning is a necessary step for ensuring the quality and reliability of hyperspectral imagery analysis for plastic waste detection. This study uses only five spectral bands essential for accurate plastic identification: Near Infrared (NIR) - 840 nm, Red Edge - 717 nm, Red - 668 nm, Green - 560 nm, and Blue bands - 475 nm. These bands are selected for their specific characteristics in distinguishing plastics, with a ±5 nm buffer for the spectral range to accommodate for imaging variability and errors due to noise.

The NIR band is useful due to plastics' distinctive absorption features in this range, while the Red Edge band captures transitional spectral variations between the Red and NIR regions. The Red, Green, and Blue bands which are part of the visible spectrum, provide complementary contrast for material differentiation.

Although the dataset available at [Ref No,] was pre-cleaned. This refinement was achieved using Quantum Geographic Information System (QGIS), an open-source tool for geospatial data analysis. QGIS in built functions are used for the isolation of the required spectral bands and the generation of monochromatic images for the selected wavelengths. The pre-processed TIFF files are used for precise pixel-level analysis and index calculations. The data was further organized into manageable formats, such as Excel sheets, enabling seamless integration with analytical tools and algorithms.

4 Proposed System

The proposed system represents a pioneering a novel solution for the global challenge of plastic waste segregation, using hyperspectral image bands and image classification techniques using spectral indices using different band imagery. This system ensures efficient separation of recyclable plastics while it can also manage detection of non-recyclable plastics which is responsible for environmental damage.

Reflectance indices serve to classify both plastic and non-plastic materials in plastic identification. Out of the 111 indices accessible for material classification, seven discriminating indices—Browning Reflectance Index (BRI), Coloration Index (CI), Intensity Index, Normalized Green Reflectance Index, Normalized Near Infrared Index (N-NIR), Normalized Red Reflectance Index, and Red Edged Index (RE) are used. These indices were selected for their dependability and discriminative potential to differentiate polymers under many environmental and imaging conditions.

The system uses data captured by the Specim FX10 hyper-spectral camera, which spans 224 wavelength bands in the visible to near-infrared spectrum [ref]. From this range, five critical central wavelengths namely: NIR (840 nm), Red Edge (717 nm), Red (668 nm), Green (560 nm), and Blue (475 nm)—are identified, with a range of ±10 nm for variability and a ±5 nm error margin. These wavelengths provide robust and precise spectral data for plastic detection

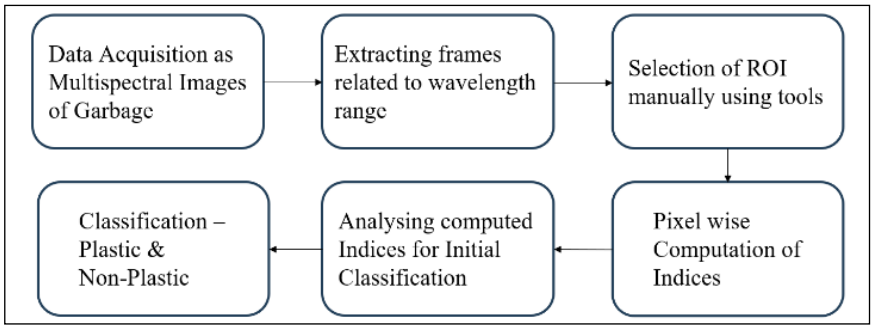

Fig.1.2 Proposed Methodology

Fig.1.2 illustrates the proposed methodology using a hyperspectral image of HDPE_BOTTLES, as a case study. A Region of Interest (ROI) is defined, and pixel-level spectral data corresponding to the selected wavelengths are extracted using library functions of Quantum Geographic Information System (QGIS). The QGIS function used to convert spectral data into monochromatic images for each wavelength, saved in TIFF format. The pixel-level intensity values are then saved into CSV format as pre-processed data for further analysis.

Using this preprocessed data, our proposed method computes the seven indices for each pixel of ROI using equation number eq.1.1 to eq.1.7 as stated above, creating a granular dataset that feeds into a proposed classification algorithm. The proposed algorithm integrates insights from all indices by correlating them to determine whether a pixel corresponds to plastic or non-plastic material, increasing the variability in material properties and imaging conditions, using the seven indices values as mentioned above

Fig.1.3 image of HDPE_BOTTLES

The reliability using Hyperspectral Vision captures spectral changes over the chosen spectral bands of wavelengths, which allows the high precision in differentiating plastics from other materials. Data processing combines computer analysis with QGIS and CSV data files. Beyond the classification of plastic waste, the detailed spectrum analysis features of the proposed system allow for study on material composition and behaviour.

4.1 Spectral Indices for Plastic Classification

Among 111 indices available for material classification provided in [1], the proposed system focuses on seven effective indices:

i. Browning Reflectance Index (BRI), eq.1.1 BRI = ((1 / G) – (1 / RE)) / NIR

ii. Coloration Index (CI), eq.1.2 CI = (R - B) / R

iii. Intensity Index, eq.1.3 Intensity = (1 / 30.5) * (R + G + B)

iv. Normalized Green Reflectance Index, eq.1.4 Normalized Green = G / (R + G + NIR)

v. Normalized Near Infrared Index (N-NIR), eq.1.5 Normalized NIR = NIR / (R + G + NIR)

vi. Normalized Red Reflectance Index, eq.1.6 Normalized Red = R / (R + G + NIR)

vii. Red Edge Index (RE), eq.1.7 RE = (NIR - R) / (NIR + R)

4.2 Pixel-Level Classification Algorithm

For each pixel within the Region of Interest (ROI), the algorithm sequentially evaluates seven spectral indices against predefined thresholds.

Pseudocode for Classification

Input

1. CSV Data of indices: Browning Reflectance Index (BRI), Coloration Index (CI), Intensity Index, Normalized Green Reflectance, Normalized NIR Reflectance, Normalized Red Reflectance, Red Edge Index (RE)
2. Pre-processed Data extracted for min and max values for each index
3. Data extracted for region of interest (ROI) dimensions: start_row, end_row, start_col, end_col

Output - Result array with classified values: PLASTIC_INDEX (1) or NON_PLASTIC_INDEX (0)

```
Initialize
- PLASTIC_INDEX = 1
- NON_PLASTIC_INDEX = 0
- result: 2D array of size ROI, initialized to zeros

For each pixel in ROI (row i from start_row to end_row, column j
from start_col to end_col):
   Initialize:
  - plastic_counter = 0
  - non_plastic_counter = 0
  For each index in indices (bri, ci, intensity, normalized_green, normalized_nir,
normalized_red, re):
      Retrieve:
      - pixel_value = value at (i, j) from the corresponding data array
      - index_range = min and max values for the current index from the data
dictionary
      If pixel_value is within index_range:
         Increment plastic_counter
      Else:
         Increment non_plastic_counter
   If plastic_counter > non_plastic_counter:
      result[i, j] = PLASTIC_INDEX
   Else:
      result[i, j] = NON_PLASTIC_INDEX
End
```

Generate monochromatic image:

- Use result array as input
- Map PLASTIC_INDEX (1) to black and NON_PLASTIC_INDEX (0) to white
- Display the resulting image

The pseudocode provides the proposed algorithm for classifying pixels within a given Region of Interest (ROI) as plastic or non-plastic. The algorithm begins by initializing constants for classification as PLASTIC_INDEX (assigned a value of 1) and NON_PLASTIC_INDEX (assigned a value of 0). It also creates a two-dimensional result array, sized according to the ROI dimensions with all initial values set to zero representing non-plastic. For each pixel within the specified ROI detected by the start and end of rows and columns, the algorithm initializes counters for tracking how many indices classify the pixel as plastic or non-plastic.

The proposed algorithm then iterates over a list of indices which are mentioned as bri (Browning Reflectance Index), ci (Coloration Index), intensity, normalized_green, normalized_nir, normalized_red, and re (Red Edge). For each index, the algorithm retrieves the pixel value from the corresponding data array and compares it against the permissible range (min and max values) specified in the data dictionary for that index. If the pixel value falls within this range, then the plastic_counter is incremented otherwise the non_plastic_counter is incremented.

After evaluating all indices for a pixel, the algorithm determines its classification based on which counter has a higher value. If the plastic_counter exceeds the non_plastic_counter, the pixel is classified as plastic pixel and the corresponding value in the result array is updated to 1 Otherwise, the pixel is classified as non-plastic pixel and the value remains 0.

Once all pixels in the ROI have been processed, the system generates a monochromatic image using the result array. In this image the pixels classified as plastic are mapped to black (value 1), while non-plastic pixels are mapped to white (value 0). The resulting image visually represents the classification, highlighting plastic regions within the ROI and facilitating further analysis. This efficient pixel-wise classification approach ensures precise identification of plastic materials in the dataset.

5 Results and Discussion

Using seven spectral indices as Browning Reflectance Index (BRI), Coloration Index (CI), Intensity, Normalized Green, Normalized Near Infrared (NIR), Normalized Red, and Red Edge (RE)—the suggested method classifies pixels in an image. These indices reflect particular spectral properties essential for plastic identification by pixel-by- pixel computation using the data of the hyperspectral imaging bands. As classification thresholds, a user-defined data dictionary offers the permissible range (minimal and maximum values) for every index. User-defined coordinates determine the Region of Interest (ROI), therefore isolating a certain portion of the image for targeted processing and so optimizing computational efficiency by pertinent areas of the image.

For every pixel in the ROI, the algorithm ranks its value against the specified thresholds for all seven spectral indices in turn. A pixel is said to help plastic categorization by incrementing the plastic_counter variable when its value for a given index falls within the derived range by empirical analysis. Values outside the range increase the non_plastic_counter variable, on the other hand. This categorization system suggests a majority- Vote reasoning: the pixel is labelled as plastic if the plastic_counter surpasses the non_plastic_counter following all indices evaluation. Otherwise, it is labelled as non-plastic. This methodical analysis guarantees that the classification uses several spectral characteristics, so improving general accuracy.

The classification process generates a result array in which every pixel is given a value of either 0 (meaning non-plastic) or 1 (showing plastic). A monochrome binary image is produced from this set. Pixels categorized as plastic are shown in black in this representation; non-plastic pixels are shown in white. The final picture offers a transparent and understandable graphic depiction of the plastic material distribution inside the ROI.

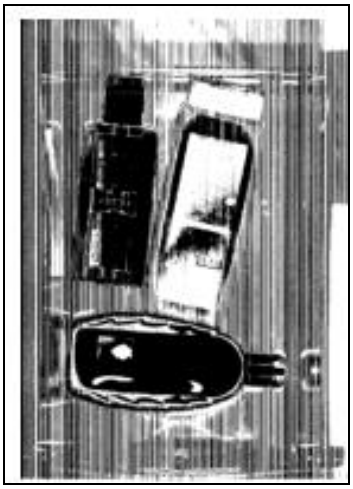

Fig.1.4 image of HDPE_BOTTLES results

5.1 Results Interpretation

5.1.1 Image Transformation

The first image in **Fig.1.3** shows the original hyperspectral dataset while the Region of Interest (ROI) is clearly marked for analysis. This provides the selected specific area from the bigger dataset background for close inspection context as a visual aid. Conversely, the second image in **Fig.1.4** illustrates a monochrome classification result derived from analysis. Black parts in this binary representation correspond to sections marked as plastic materials, white areas indicate non-plastic materials.

5.1.2 Key Insights

The system of classification demonstrates great ability to divide plastic materials inside complex waste situations. Its pixel-level grain guarantees the detection of even little or partially covered plastic objects, therefore displaying its extraordinary sensitivity and efficiency. Such accuracy is quite important for pragmatic applications in waste segregation and recycling, where the correct identification of plastics is crucial to maximize the recycling process.

5.1.3 Technical Efficacy

By use of seven spectral indices, we can largely classify the spectral features unique to plastics. The classification accuracy is significantly increased and the potential of classification errors is much reduced by depending on a multi-index technique rather than a single index. Changing parameters including the ROI and threshold settings will help one control larger datasets and adapt to match various imaging conditions. This flexibility qualifies the system for numerous applications, including waste management on industrial level and environmental monitoring.

6 Conclusion and Future Scope

This study introduces a systematic approach utilizing machine learning techniques and specific spectral features from hyper-spectral images for the segregation of plastic waste. The proposed approach utilizes spectral indices and data-driven algorithms to

detect and isolate plastic waste with increasing accuracy. The method utilizes hyperspectral imagery and seven fundamental spectral indices to accurately identify plastic components within complex waste environments. The system processes pixel-level spectral data within specified regions of interest (ROI) to generate monochromatic images, where black regions represent plastics and white regions denote non-plastic materials. In addition to addressing hand sorting inefficiencies, this comprehensive classification ensures precise identification of recyclable plastics, thereby enhancing the overall recycling process.

Integrating mid-infrared (MIR) and far-infrared (FIR) regions, along with advanced spectral unmixing techniques, can enhance the understanding of sub-pixel compositions in mixed materials, thereby facilitating the identification of materials with overlapping characteristics, such as LDPE and HDPE. The dataset generated through this approach may serve as a valuable resource for researchers and policymakers, thereby facilitating data-driven waste management decisions.

References

1. Iordache, M.-D.; De Keukelaere, L.; Moelans, R.; Landuyt, L.; Moshtaghi, M.; Corradi, P.; Knaeps, E. Targeting Plastics: Machine Learning Applied to Litter Detection in Aerial Multispectral Images. Remote Sens. 2022, 14, 5820. https://doi.org/10.3390/rs14225820
2. Tasseron, P.; van Emmerik, T.; Peller, J.; Schreyers, L.; Biermann, L. Advancing Floating Macroplastic Detection from Space Using Experimental Hyperspectral Imagery. Remote Sens. 2021, 13, 2335. https://doi.org/10.3390/rs13122335
3. Abraham Vázquez-Guardado, Mason Money, Nathaniel McKinney, and Debashis Chanda, "Multi-spectral infrared spectroscopy for robust plastic identification," Appl. Opt. 54, 7396-7405 (2015)
4. Cortesi, Irene & Masiero, Andrea & Tucci, Grazia & Topouzelis, K.. (2022). UAV-BASED RIVER PLASTIC DETECTION WITH A MULTISPECTRAL CAMERA. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences. XLIII-B3-2022. 855-861. 10.5194/isprs-archives-XLIII-B3-2022-855-2022
5. Henriksen, Martin & Karlsen, Celine & Klarskov, Pernille & Hinge, Mogens. (2021). Plastic Classification via In-line Hyperspectral Camera Analysis and Unsupervised Machine Learning. Vibrational Spectroscopy. 118. 103329. 10.1016/j.vibspec.2021.103329
6. Harmse, C.J.; Gerber, H.; van Niekerk, A. Evaluating Several Vegetation Indices Derived from Sentinel-2 Imagery for Quantifying Localized Overgrazing in a Semi-Arid Region of South Africa. Remote Sens. 2022, 14, 1720. https://doi.org/10.3390/rs14071720
7. R. Mayer, F. Bucholtz and D. Scribner, "Object detection by using "whitening / de-whitening" to transform target signatures in multitemporal hyperspectral and multispectral imagery," in IEEE Transactions on Geoscience and Remote Sensing, vol. 41, no. 5, pp. 1136-1142, May 2003, doi: 10.1109/TGRS.2003.813553
8. S. Song et al., "Deep Domain Adaptation Based Multi-Spectral Salient Object Detection," in IEEE Transactions on Multimedia, vol. 24, pp. 128-140, 2022, doi: 10.1109/TMM.2020.3046868
9. Gani, M.O., Kuiry, S., Das, A., Nasipuri, M., Das, N. (2021) “Multispectral Object Detection with Deep Learning” In: Dutta, P., Mandal, J.K., Mukhopadhyay, S. (eds) Computational Intelligence in Communications and Business Analytics. CICBA 2021. Communications in Computer and Information Science, vol 1406. Springer, Cham. https://doi.org/10.1007/978-3-030-75529-4_9
10. A. Pramanik, S. K. Pal, J. Maiti and P. Mitra, "Granulated RCNN and Multi-Class Deep SORT for Multi-Object Detection and Tracking," in IEEE Transactions on Emerging Topics in Computational Intelligence, vol. 6, no. 1, pp. 171-181, Feb. 2022, doi: 10.1109/TETCI.2020.3041019.

11. Y. Wang et al., "Supervised High-Level Feature Learning with Label Consistencies for Object Recognition," in IEEE Transactions on Geoscience and Remote Sensing, vol. 58, no. 7, pp. 4501-4516, July 2020, doi: 10.1109/TGRS.2019.2955557.
12. C. Liu, D. Q. Huynh, Y. Sun, M. Reynolds and S. Atkinson, "A Vision-Based Pipeline for Vehicle Counting, Speed Estimation, and Classification," in IEEE Transactions on Intelligent Transportation Systems, vol. 22, no. 12, pp. 7547-7560, Dec. 2021, doi: 10.1109/TITS.2020.3004066.
13. Janusz Bobulski, Mariusz Kubanek, "Deep Learning for Plastic Waste Classification System", Applied Computational Intelligence and Soft Computing, vol. 2021, Article ID 6626948, 7 pages, 2021. https://doi.org/10.1155/2021/6626948
14. Raju, K & Thiagarajan, Meera Devi. (2021). "Automatic Plastic Waste Segregation and Sorting Using Deep Learning Model" In International Journal of Scientific & Technology Research Volume 9, Issue 02, February 2020
15. Shichao Zhu, Honghui Chen, Mengmeng Wang, Xuemei Guo, Yu Lei, Gang Jin "Plastic solid waste identification system based on near infrared spectroscopy in combination with support vector machine" in Advanced Industrial and Engineering Polymer Research Volume 2, Issue 2, April 2019, Pages 77-81 https://doi.org/10.1016/j.aiepr.2019.04.001
16. I. W. Raka Ardana, I. Bagus Irawan Purnama and I. M. Sumerta Yasa, "Application of Object Recognition for Plastic Waste Detection and Classification Using YOLOv3," 2020 International Conference on Applied Science and Technology (iCAST), 2020, pp. 652-656, doi: 10.1109/iCAST51016.2020.9557735.
17. L. R. Kambam and R. Aarthi, "Classification of plastic bottles based on visual and physical features for waste management," 2019 IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT), 2019, pp. 1-6, doi: 10.1109/ICECCT.2019.8869191.
18. Ziouzios, D.; Tsiktsiris, D.; Baras, N.; Dasygenis, "M. A Distributed Architecture for Smart Recycling Using Machine Learning". Future Internet 2020, 12, 141. https://doi.org/10.3390/fi12090141